KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

YEAR 10 TRANSITION · 2026

# Year 10 Mathematics Yearly Test

NAME _____ CLASS _____

TEACHER _____ DATE _____

| WORKING TIME | TOTAL MARKS | CALCULATORS | BRING |
|---|---|---|---|
| **1 hour 15 min** | **55 marks** | **Allowed** | **Pen, pencil** |
| 75 minutes | 35 questions | bring your own | ruler, eraser, calculator |

## WHAT IS IN THIS TEST

| | Section and topics | Questions | Marks | Time | Marker use |
|---|---|---|---|---|---|
| A | **Financial mathematics**<br>Percentages, GST, profit, discount, wages, loans, interest | 1–11 | **15** | 20 min | _____ / 15 |
| B | **Measurement**<br>Units, perimeter, circles, area, surface area, volume | 12–21 | **15** | 20 min | _____ / 15 |
| C | **Statistics and probability**<br>Types of data, stem-and-leaf, mean, median, graphs, probability | 22–29 | **14** | 20 min | _____ / 14 |
| D | **Other Year 10 topics**<br>Index laws, algebra, equations, straight-line graphs, Pythagoras, trigonometry | 30–35 | **11** | 15 min | _____ / 11 |
| | **Total** | **1–35** | **55** | **75 min** | _____ / 55 |

## INSTRUCTIONS

1. Use a **blue or black pen.** Use a pencil and ruler for graphs and diagrams.
2. **Answer every question.** Write your answer on the **Answer** line, or in the table, box or graph.
3. **Show your working.** You can get marks for working, even if your answer is wrong.
4. **Multiple choice:** circle **one** letter, A, B, C or D.
5. Watch the clock. Each section has a **suggested time.**

## STUCK? LOOK FOR HELP

Many questions have a **HELP** box. It gives you the **formula** and a worked **example**, like this:

**HELP**

**Formula** Speed = distance ÷ time

**Example** 100 km in 2 hours:
100 ÷ 2 = 50 km/h

**Do not turn over until your teacher tells you to start. Good luck!**

SECTION A

# Financial mathematics

Questions **1–11** · **15 marks** · Suggested time **20 minutes**

____ / 15
Marker

Questions 1–4 are **multiple choice**. Circle the letter of the correct answer, like this: (B)

**1** What is 28% of $89? **1 mark**

(A) $28.00 (B) $24.92 (C) $64.08 (D) $113.92

**HELP** **Example** 10% of $50:
0.10 × 50 = $5

**2** A price tag says "**$55 (GST inc.)**". How much GST is included in the price? **1 mark**

(A) $5.00 (B) $5.50 (C) $50.00 (D) $60.50

**HELP** **Remember** GST in a price (GST inc.) = price ÷ 11

**3** A shirt costs $56. It is **discounted by 26%**. What is the new price? **1 mark**

(A) $14.56 (B) $26.56 (C) $30.00 (D) $41.44

**HELP** **Remember** Find 26% of $56, then **take it off** the price.

**4** Find the **simple interest** on $600 at 5% per year for 4 years. **1 mark**

(A) $12.00 (B) $30.00 (C) $120.00 (D) $720.00

**HELP** **Formula** $I = P \times R \times N$
(5% → 0.05)

**5** Increase $968 by 12%. Use a multiplier and your calculator. **1 mark**

Working

**HELP**

**Multiplier** Increase by 8% → × 1.08
Decrease by 8% → × 0.92

**Example** Increase $50 by 10%:
50 × 1.10 = $55

**Answer** $ _____


Section A continues on the next page →

**6** A shop buys a cap for \$7.60. This is the **cost price**. The shop adds a **50% mark-up**. **2 marks**

**a** Find the selling price of the cap.

Working

**Answer** \$ _____

**b** How much profit does the shop make?

Working

**Answer** \$ _____

**HELP**

**Remember** Mark-up of 30% → × 1.3
Profit = selling price − cost price

**Example** Cost \$10, mark-up 20%: 10 × 1.2 = \$12, profit = \$2

**7** Tom is paid \$15.78 per hour. **2 marks**

**a** Tom works 30 hours. How much is he paid?

Working

**Answer** \$ _____

**b** Tom also works 4 hours of overtime at **time and a half**. How much is he paid for the overtime?

Working

**Answer** \$ _____

**HELP**

**Remember** Pay = hours × hourly rate
Time and a half = 1.5 × hourly rate

**Example** \$20 per hour, 3 hours at time and a half: 3 × 1.5 × 20 = \$90

**8** Pedro sells phones. He earns **4.5% commission** on his sales. His sales this week are \$790. How much commission does he earn? **1 mark**

Working

**Answer** \$ _____

**HELP**

**Remember** 4.5% → 0.045
Commission = 0.045 × sales

**9** Tranh works 36 hours at $28.89 per hour. **2 marks**

**a** Find Tranh's **gross pay** (total pay before anything is taken out).

Working

Answer $ _____

**b** Tranh pays $142.59 in tax and $22.50 in super (superannuation). Find his **net pay** (the pay he takes home).

Working

Answer $ _____

**HELP**

**Remember** Gross pay = hours × hourly rate
Net pay = gross pay − tax − super

**10** Mia buys a phone on a payment plan. She borrows $1 200 and repays **$60 a month for 2 years**. How much **interest** does she pay? **1 mark**

Working

Answer $ _____

**HELP**

**Remember** 2 years = 24 months
Total repaid = repayment × number of months
Interest = total repaid − amount borrowed

**Example** Borrow $500, repay $50 a month for 1 year:
50 × 12 = $600
Interest = 600 − 500 = $100

**11** $5 000 is invested for 3 years. It earns 4% **compound interest** each year. **2 marks**

**a** Find the value of the investment after 3 years.

Working

Answer $ _____

**b** How much interest is earned?

Working

Answer $ _____

**HELP**

**Formula** $A = P(1 + R)^n$
$P$ = amount invested, $R$ = rate as a decimal, $n$ = number of years

**Remember** 5% → 1 + 0.05 = 1.05

**Example** $2 000 at 5% for 2 years:
$A = 2000 \times 1.05^2 = \$2\,205$
Interest = 2205 − 2000 = $205

**Calculator** Type 2000 × 1.05 ^ 2 =

SECTION B

# Measurement

Questions **12–21** · **15 marks** · Suggested time **20 minutes**

_____ / 15
Marker

Questions 12–15 are **multiple choice**. Circle the letter of the correct answer, like this: (B)

**12** How many metres are in 1 kilometre? **1 mark**

(A) 10 (B) 100 (C) 1 000 (D) 10 000

**13** What is the **area** of a rectangle 3 cm long and 4 cm wide? **1 mark**

(A) 7 $cm^2$ (B) 12 cm (C) 14 cm (D) 12 $cm^2$

**14** What is the **volume** of a cube with sides 3 cm long? **1 mark**

(A) 9 $cm^3$ (B) 27 $cm^3$ (C) 27 $cm^2$ (D) 54 $cm^3$

**HELP** **Formula** Volume of a cube = side × side × side

**15** Which of these is 232 000 written in **scientific notation**? **1 mark**

(A) $2.32 \times 10^5$ (B) $23.2 \times 10^4$

(C) $2.32 \times 10^{-5}$ (D) $232 \times 10^3$

**HELP** **Example** $2\,350\,000 = 2.35 \times 10^6$ (one digit before the decimal point)

**16** Convert each measurement. **2 marks**

**a** Change 5.3 km into metres.

**Answer** _____ m

**b** Change 2.5 L into millilitres.

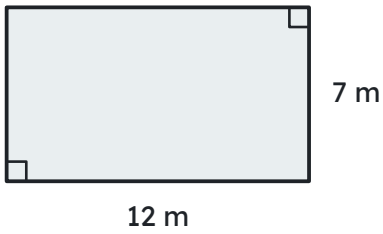
**Answer** _____ mL

**HELP**

**Remember** km → m: × 1000
L → mL: × 1000

**17** A farmer puts a fence all the way around this rectangular paddock. How many metres of fence are needed? **1 mark**

**HELP**
**Remember** Perimeter = add up the lengths of **all** the sides.




Working

**Answer** _____ m

**18** This circle has a radius of 3 m. Give your answers to 1 decimal place. **2 marks**

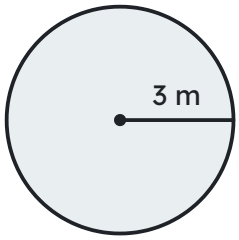
**a** Find the **circumference**.

Working

**Answer** _____ m

**HELP**
**Formula** $C = 2 \times \pi \times r$ (circumference)
$A = \pi \times r \times r$ (area)
**Example** Radius 5 cm:
$C = 2 \times \pi \times 5 = 31.4$ cm
$A = \pi \times 5 \times 5 = 78.5$ cm$^2$

**b** Find the **area**.

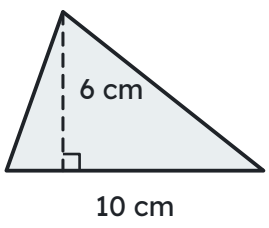



Working

**Answer** _____ m$^2$

**19** Find the area of each shape. **2 marks**

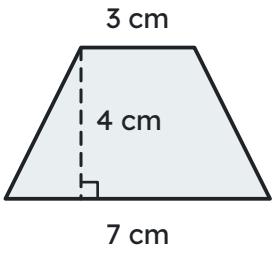
**a** Triangle




Working

**Answer** _____ cm$^2$

**b** Trapezium

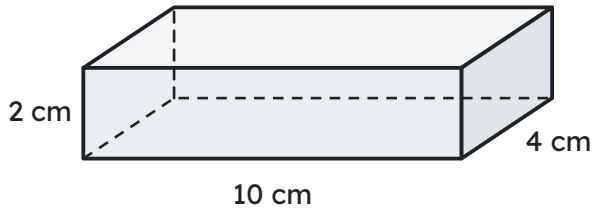



Working

**Answer** _____ cm$^2$

**HELP**
**Formula** Triangle: $A = ½ \times$ base $\times$ height
Trapezium: $A = ½ \times$ (top + bottom) $\times$ height
(top and bottom are the parallel sides)
**Example** Triangle, base 8 cm, height 5 cm:
$½ \times 8 \times 5 = 20$ cm$^2$

**20** Find the **surface area** of this box. Complete the table. The first row is done for you. **2 marks**




**HELP**

**Remember** A box has 6 faces, in 3 matching pairs. Surface area = add the areas of all 6 faces.

| Faces | Area of one face | Number of faces | Area |
|---|---|---|---|
| **Top and bottom** | $10 \times 4 = 40$ cm$^2$ | 2 | 80 cm$^2$ |
| **Front and back** | | 2 | |
| **The two ends** | | 2 | |
| **Surface area (add the last column)** | | | |

**21** This cylinder has a radius of 2 cm and a height of 6 cm. **2 marks**

**a** Find the **volume**, to 1 decimal place.

Working

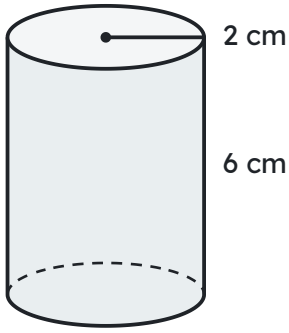
**HELP**

**Formula** Volume of a cylinder: $V = \pi \times r \times r \times h$

**Example** $r = 3$ cm, $h = 10$ cm: $V = \pi \times 3 \times 3 \times 10 = 282.7$ cm$^3$

**Remember** 1 cm$^3$ holds 1 mL

**Answer** _____ cm$^3$




**b** How many millilitres (mL) of water can it hold?

Working

**Answer** _____ mL

SECTION C

# Statistics and probability

Questions **22–29** · **14 marks** · Suggested time **20 minutes**

____ / 14
Marker

Questions 22–25 are **multiple choice**. Circle the letter of the correct answer, like this: (B)

**22** A survey asks: **"What is your favourite sport to play?"** What type of data does this give? **1 mark**

(A) numerical and discrete

(B) numerical and continuous

(C) categorical and ordinal

(D) categorical and nominal

**HELP**

**Remember** Categorical = words or groups. Numerical = numbers.
Nominal = no order (eye colour, type of pet).
Ordinal = has an order (small, medium, large).

Use this **stem-and-leaf plot** for Questions **23** and **24**.
It shows the scores of 11 players in a game.

| Stem | Leaf |
|---|---|
| 2 | 4 9 |
| 3 | 1 1 7 8 |
| 4 | 2 4 6 |
| 5 | 0 4 |

4 | 2 means 42

**23** What is the **lowest** score? **1 mark**

(A) 4 (B) 20 (C) 24 (D) 54

**24** What is the **mode** (the most common score)? **1 mark**

(A) 3 (B) 31 (C) 38 (D) 42

**25** A normal 6-sided dice is rolled once. What is the probability of rolling an **even** number? **1 mark**

(A) $\frac{1}{6}$ (B) $\frac{1}{3}$ (C) $\frac{1}{2}$ (D) $\frac{2}{3}$

**HELP** **Formula** Probability = number of ways it can happen ÷ total number of ways


Section C continues on the next page →

**26** These are the goals a netball player scored in 7 games: **3 marks**

**8, 3, 13, 1, 5, 3, 9**

Write the numbers in order first: ☐ ☐ ☐ ☐ ☐ ☐ ☐

**a** Find the **mean**.

Working

**Answer** _____

**b** Find the **median**. **Answer** _____

**c** Find the **range**. **Answer** _____

**HELP**

| | |
|---|---|
| **Mean** | Add up the numbers, then divide by how many numbers there are |
| **Median** | The middle number, when they are in order |
| **Range** | Biggest – smallest |

**27** One letter is chosen at random from the word **2 marks**

| I | N | T | E | R | E | S | T |
|---|---|---|---|---|---|---|---|

**a** Find the probability that the letter is an **E**.

**Answer** _____

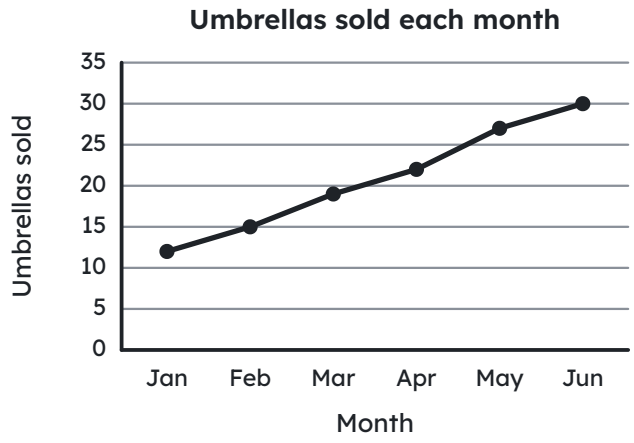
**b** Find the probability that the letter is a **vowel** (A, E, I, O or U).

**Answer** _____

**HELP**

| | |
|---|---|
| **Formula** | Probability = number of ways it can happen ÷ total number of ways |
| **Example** | A letter from **BOOK**: $P(O) = \frac{2}{4} = \frac{1}{2}$ |

**28** This **time-series graph** shows how many umbrellas a shop sold each month. **1 mark**




**HELP**

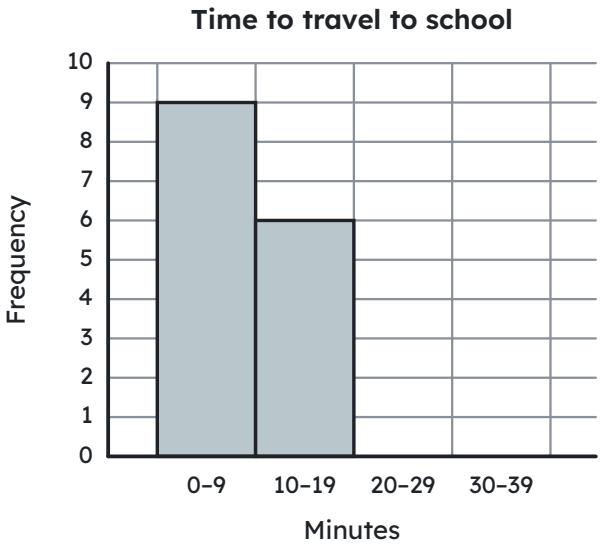
| | |
|---|---|
| **Remember** | Upward trend: the line goes up over time. Downward trend: the line goes down over time. |

Describe the **trend**. Circle one.

**upward trend** **downward trend** **no trend**

**29** Some students recorded how many minutes it takes them to travel to school. **4 marks**

| Minutes | Frequency |
|---|---|
| 0–9 | 9 |
| 10–19 | 6 |
| 20–29 | 3 |
| 30–39 | 2 |
| **Total** | |




**a** Fill in the **total** at the bottom of the table.

**b** The graph has 2 columns drawn. Use the table to **draw the columns** for 20–29 and 30–39 on the graph.

**c** What **percentage** of the students took 0–9 minutes?

Working

**HELP**

**Formula** Percentage = frequency ÷ total × 100

**Shapes**




**Answer** _____ %

**d** Is the data **symmetrical** or **skewed**? Circle one.

symmetrical skewed

SECTION D

# Other Year 10 topics

Questions **30–35** · **11 marks** · Suggested time **15 minutes**

_____ / 11
Marker

Question 30 is **multiple choice**. Circle the letter of the correct answer, like this: (B)

**30** Simplify $a^3 \times a^4$ — **1 mark**

(A) $a^7$ (B) $a^{12}$ (C) $2a^7$ (D) $7a$

**HELP** **Remember** When you multiply, **add** the powers. $m^2 \times m^5 = m^7$

**31** If $a = 4$, find the value of $5a - 2$. — **1 mark**

Working

**HELP**
**Remember** $5a$ means $5 \times a$
**Example** If $a = 3$:
$2a + 1 = 2 \times 3 + 1 = 7$

**Answer** _____

**32** Solve each equation. — **2 marks**

**a** $2x + 3 = 13$

Working

**Answer** $x =$ _____

**HELP**
**Remember** Do the **opposite** to both sides.
**Example** $3x + 1 = 13$
Take away 1: $3x = 12$
Divide by 3: $x = 4$
**Example** $\frac{x+1}{2} = 4$
Times by 2: $x + 1 = 8$
Take away 1: $x = 7$

**b** $\frac{x+2}{3} = 4$

Working

**Answer** $x =$ _____


Section D continues on the next page →

**33** The rule for a straight line is $y = 2x + 1$. **3 marks**

**a** Use the rule to find $y$ when $x = 0$, 1, 2 and 3. Write each answer in the table. 1 mark

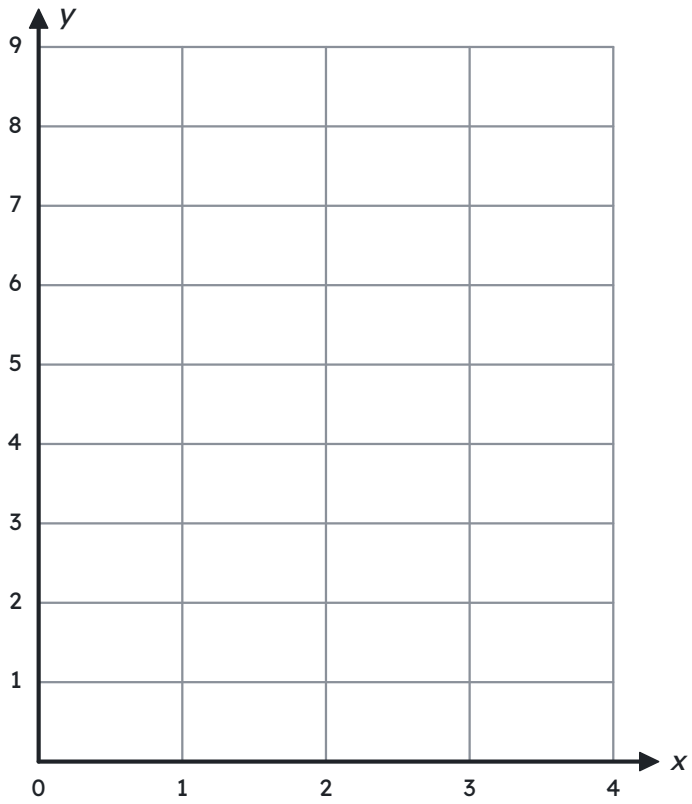
| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $y$ | | | | |

**b** Plot the points on the grid and join them with a straight line. 1 mark

**c** Write the **gradient** and the ***y*-intercept** of $y = 2x + 1$. 1 mark

Gradient = _____

*y*-intercept = _____




**HELP**

**Example** For $y = 3x + 2$, when $x = 4$: $y = 3 \times 4 + 2 = 14$

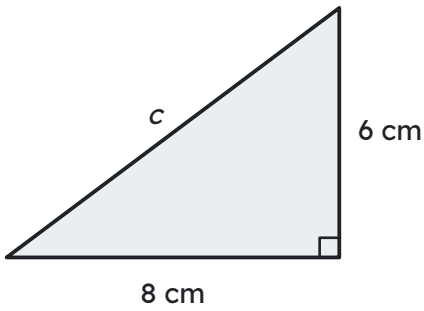
**Plot** For $x = 1$ and $y = 3$: go across to 1, then up to 3. Put a dot. Join the dots with a ruler.

**$y = mx + c$** $m$ is the gradient (how steep the line is). $c$ is the $y$-intercept (where the line crosses the $y$-axis).

**34** Find the length of the **hypotenuse**, *c*.

**2 marks**

Working




**HELP**

**Formula** $c^2 = a^2 + b^2$ ($c$ is the longest side)

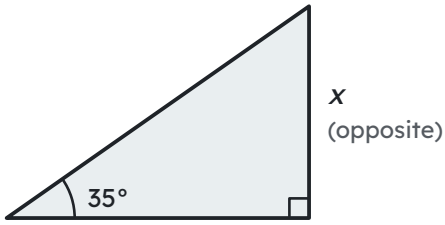
**Example** $a = 3$, $b = 4$:
$c^2 = 9 + 16 = 25$
$c = \sqrt{25} = 5$

**Answer** $c =$ _____ cm

**35** Find the length of side *x*, to 1 decimal place.

**2 marks**

Working




**HELP**

**Formula** tan (angle) = opposite ÷ adjacent
so opposite = adjacent × tan (angle)

**Example** Angle 20°, adjacent 5 cm:
$x = 5 \times \tan 20° = 1.8$ cm

**Calculator** Check your calculator shows **D** or **DEG** (degrees).

**Answer** $x =$ _____ cm

## End of the test

- Did you answer **every** question? Have a go at any you missed.
- Did you write the **units** ($, cm, $m^2$, mL) and show your **working**?
- Use the space below if you need more room. Write the question number next to your working.

Extra working space